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Research on a Universal Analytical Thermal Circuit Model for Civil Electric Cables

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Abstract

The measured standard resistance at 20 °C (R_{20}) is a critical indicator for evaluating the quality of electric cables. Utilizing a thermal circuit model allows for the rapid determination of R_{20}. Focusing on civil electrical cables, this study constructs a thermal circuit model based on an equivalent circuit. Through mathematical derivation, model parameters are expressed as functions of the insulation cross-sectional area and conductor specification. Subsequently, the insulation area is fitted to the conductor specification, establishing a universal analytical thermal circuit model with the conductor specification as the sole variable. Ambient temperatures measured under specific operating conditions showed exponential variation. Using corresponding simulated conductor temperatures, the Differential Evolution algorithm was employed to train the model, achieving a training standard deviation of 0.0184 °C. Validation under different conditions demonstrated that for various cable specifications, the conductor temperature prediction deviation within 300 s remained within 0.368 °C, and the maximum estimation error for R_{20} was less than 0.148%. These results indicate that the established model possesses high calculation accuracy and strong universality, offering a valuable tool for researchers and practitioners in fields relevant to civil electrical cables.

Keywords: electric cables; resistance; temperature; analytical model

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Academic Editor: Armando Oliveira

Received: 12 December 2025
Revised: 24 December 2025
Accepted: 29 December 2025
Published: 31 December 2025

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1. Introduction

Electric cables are indispensable infrastructure components in modern industries and power grids. Accurately assessing their quality and health status is pivotal for ensuring the safe and reliable operation of power systems. Among the various assessment parameters, the standard conductor resistance at 20 °C (R_{20}) serves as a critical indicator that reflects the overall quality and health conditions, such as aging and corrosion, of the cables. Therefore, the precise measurement of R_{20} lays the foundation for power system analysis.

However, due to the sensitivity of electrical resistance to temperature, even minor deviations in conductor temperature can lead to inaccurate measurement results [1–4]. Consequently, the R_{20} measurement protocols stipulated by the International Electrotechnical Commission (IEC) [5] and the Chinese National Standard (GB) [6] mandate that the cable sample be placed in a temperature-stable environment. The conductor resistance can only be measured after the conductor temperature has reached equilibrium with the ambient temperature. This equilibration process requires a waiting period of a substantial duration,

resulting in a prolonged testing cycle and low efficiency. However, given the established conversion relationship between the measured resistance R_t at a specific temperature t and the standard resistance R_{20}, this lengthy waiting period can be eliminated. Specifically, if the real-time conductor temperature t can be obtained, R_t can be measured directly and converted to R_{20} without waiting for thermal equilibrium.

Methods for obtaining the conductor temperature of electric cables can be primarily categorized into direct measurement methods and indirect calculation methods. Direct measurement methods employ sensors to directly acquire the conductor temperature. For instance, Zhou et al. [7] utilized distributed optical fiber sensing and thermocouples to measure the temperature of submarine cables to validate an equivalent thermal circuit. Similarly, Wang et al. [8] employed thermocouples to measure the conductor temperature of tunnel cables to verify the accuracy of a transient thermal circuit model. However, direct measurement methods typically require complex measurement setups and incur high costs. Furthermore, embedding thermocouples inside the cable structure makes it difficult to guarantee measurement accuracy.

Indirect calculation methods employ mathematical models to describe the temperature dynamics of electric cables, thereby deriving the real-time conductor temperature through calculation. The Finite Element Method (FEM) is commonly used to simulate and determine the precise temperature of cables under various environmental conditions. For instance, Chen et al. [9] used COMSOL Multiphysics 6.3 software to analyze the temperature of three-core 10 kV AC XLPE cables and their joints under thermo-electric coupling conditions. Dulaimi et al. [10] utilized FEM simulations to obtain conductor temperatures of underground cables coupled with different soil conditions and cable configurations; they then used the simulated data to train an adaptive Back-Propagation Neural Network for temperature prediction. Similarly, Karahan et al. [11] obtained conductor temperatures under various environmental scenarios using FEM based on specific cable characteristics. Although FEM calculations can yield highly accurate results, they demand significant computational resources and involve long computation times [12].

The International Electrotechnical Commission has issued the IEC-60287 [13] and IEC-60853 [14] standards. These standards employ a thermo-electric analogy method to establish an analytical thermal circuit model for solving the temperature field of cables, a method that has long been utilized by power institutions. The IEC method enables the efficient calculation of cable temperature variations, requiring minimal computation time and demonstrating high computational efficiency [15]. However, Sedaghat and Francisco [16] compared the results obtained from the IEC method, the FEM, and experimental approaches. They found that because the empirical formulas in the IEC standards are often based on assumptions of ideal contact and specific boundary conditions—ignoring the effects of actual contact thermal resistance and radiative heat transfer—the resulting calculation accuracy is relatively low.

Numerous studies have sought to improve indirect calculation methods by combining the advantages of the FEM and analytical thermal circuit models. For instance, Xiao et al. [12] utilized FEM simulations to obtain temperature rise data for cables and subsequently applied a Genetic Algorithm (GA) to determine the coefficient values of the thermal circuit model based on the simulated data. Similarly, Liang et al. [17] constructed a thermal circuit model for duct cables based on the heat transfer relationships among the cable body, the air inside the duct, and the external environment; they then employed FEM simulation results to solve for the thermal resistance coefficients in the model. These improved indirect calculation methods possess the dual advantages of high accuracy and computational convenience. However, the model coefficients obtained through training are

typically applicable only to specific cables and application scenarios, making them unsuitable for conductor measurements involving multiple specifications and diverse scenarios.

Cables for civil applications are extensively used in daily life, and the measurement of their conductor resistance is a critical measure for ensuring operational safety. However, most existing studies on conductor temperature measurement focus on large-gauge high-voltage cables or overhead lines, while research on low-voltage, small-gauge cables for civil use remains scarce. Conductors in civil cables typically have cross-sectional areas not exceeding 10 mm^2 and come in a variety of specifications. Focusing on the laboratory inspection of conductor resistance for civil cables, this study employs a method combining the FEM and an analytical thermal circuit model. By constructing a thermal circuit model tailored to the characteristics of inspection applications and solving its coefficients, this research aims to develop a universal thermal circuit model applicable to various specifications of civil cables.

The remainder of this paper is organized as follows: Section 2 constructs the analytical thermal circuit model; Section 3 solves for the model coefficients; Section 4 validates the accuracy of the model; and Section 5 presents the discussion and conclusions.

2. Formulation of the Analytical Thermal Circuit Model

The structure of the wire and cable specimen used for testing is shown in Figure 1. It consists of an internal cylindrical conductor wrapped in an insulation layer. To facilitate clamping, a portion of the insulation at both ends is stripped to expose the conductor.

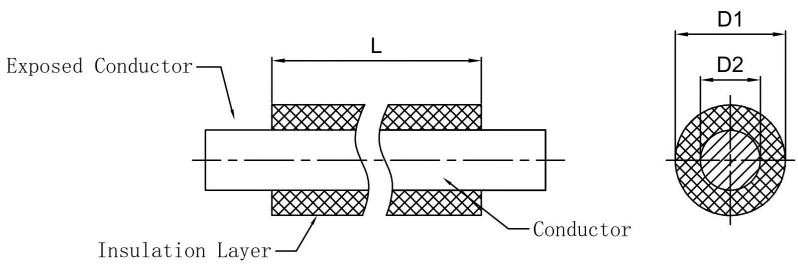


Figure 1. The structure of the cable.

2.1. Construction of the Equivalent Thermal Network

To simplify the analytical model while maintaining engineering accuracy, given the tight extrusion manufacturing process of civil electric cables, and in accordance with the calculation methods outlined in IEC 60287 [13], the interface between the conductor and insulation layer is assumed to be seamless with negligible gaps; thus, contact thermal resistance is ignored. Furthermore, as heat transfer within the solid cable structure is dominated by conduction, internal radiative heat transfer is considered negligible.

For the insulated section of the conductor, the heat transfer path is: Ambient—Insulation—Conductor. For the exposed conductor parts at both ends, the heat transfer path is: Ambient—Conductor. As illustrated in Figure 2, the wire and cable are modeled as a thermal network consisting of two thermal nodes in series: the insulation layer and the conductor. Specifically, the insulation node consists of an R_1C_1 series thermal circuit, while the conductor node consists of an R_2C_2 series thermal circuit. Here, R_1 and R_2 denote the equivalent thermal resistances of the insulation layer and the conductor, respectively, while C_1 and C_2 represent their respective equivalent thermal capacitances. The thermal power transferred through R_1, C_1, R_2, and C_2 are denoted as Q_{11}, Q_{12}, Q_{21}, and Q_{22}, respectively, and the thermal power transferred through the exposed conductor parts is denoted as Q_k.

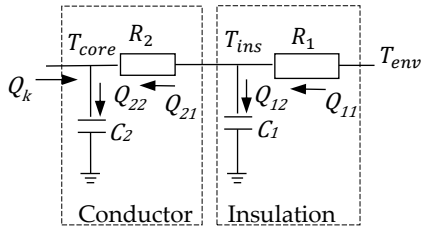


Figure 2. Equivalent thermal network of the electric cable. The arrows indicate the direction of heat flow.

Based on the relevant theory of the thermal circuit model, substituting thermal power with thermal current and temperature with thermal potential yields:

$$Q_{12} = Q_{11} - Q_{21}, \tag{1}$$

$$Q_{22} = Q_{21} + Q_k. \tag{2}$$

Here, Q_k is linearly related to the temperature difference between the conductor and the ambient environment. Consequently, the equivalent heat transfer mathematical equations for the wire and cable can be derived as:

$$\frac{dT_{ins}}{dt} = \frac{T_{env} - T_{ins}}{R_1 C_1} - \frac{T_{ins} - T_{core}}{R_2 C_1}, \tag{3}$$

$$\frac{dT_{core}}{dt} = \frac{T_{ins} - T_{core}}{R_2 C_2} + \frac{k_1 (T_{env} - T_{core})}{C_2}. \tag{4}$$

Here, T_{core} is the conductor temperature, T_{ins} is the temperature of the insulation layer, T_{env} is the ambient temperature, and k_1 is the heat transfer coefficient of the exposed end.

2.2. Construction of the Universal Thermal Circuit Model

It is evident from Equations (1) and (2) that the thermal equilibrium model is governed by five parameters: R_1, C_1, R_2, C_2, and k_1. Notably, these parameters are correlated with the geometric dimensions of the wire and cable conductor. In practical applications, the targets often encompass a series of wire products with varying specifications; consequently, the thermal equilibrium model parameters will differ for different products. If the model is calibrated for a specific product specification, the resulting parameters will be inapplicable to others. Therefore, this study attempts to establish functional relationships between the geometric dimensions of the wire and cable and the thermal equilibrium model parameters, aiming to endow the model with the universality required to handle the entire product series.

This study primarily focuses on copper conductor cables with cross-sectional areas of 10 mm^2 or less, which are extensively used in civil applications. In accordance with the national standard GB/T 5023.3-2008 [18], copper core cables within this size range ($\leq$10 mm^2) were selected as the research subjects for the construction of the universal physical parameter model.

2.2.1. Calculation Equations for Equivalent Thermal Capacitance of the Insulation Layer and Conductor

Since both the insulation layer and the conductor are cylindrical geometries, their thermal capacitance values are calculated as follows:

$$C_1 = c_{p_1} \rho_1 S_1 L, \tag{5}$$

$$C_2 = c_{p_2}\rho_2 S_2 L. \tag{6}$$

Here, c_{p_1} and c_{p_2} are the specific heat capacities of the insulation layer and the conductor, respectively; ρ_1 and ρ_2 are the densities of the insulation layer and the conductor, respectively; S_1 and S_2 are the cross-sectional areas of the insulation layer and the conductor, respectively; and L is the length of the insulation layer.

Given that the materials for the insulation layer and the conductor are fixed as polyvinyl chloride (PVC) and copper, respectively, and that the length L is fixed at 1 m in accordance with the measurement requirements for conductor resistance, the parameters c_{p1}, c_{p2}, ρ_1, ρ_2, and L are all constants. Thus, we have:

$$C_1 = k_{c1} \cdot S_1, \tag{7}$$

$$C_2 = k_{c2} \cdot S_2, \tag{8}$$

$$k_{c1} = c_{p_1}\rho_1 L, \tag{9}$$

$$k_{c2} = c_{p_2}\rho_2 L. \tag{10}$$

Here, k_{c1} and k_{c2} are the constant coefficients for the calculation of C_1 and C_2, respectively.

2.2.2. Calculation Equations for Equivalent Thermal Resistance of the Insulation Layer and Conductor

Heat exchange occurs between the insulation layer and the conductor through their outer surfaces. According to the cylindrical heat transfer formulas [8], R_1 and R_2 are expressed as:

$$R_1 = \frac{1}{hA_1}, \tag{11}$$

$$R_2 = \frac{\rho}{2\pi} ln\left(\frac{D_1}{D_2}\right). \tag{12}$$

Here, h is the combined heat transfer coefficient of the wire and cable surface; A_1 is the outer surface area of the insulation layer; D_1 and D_2 are the diameters of the insulation layer and the conductor, respectively; and ρ is the thermal resistance coefficient of the insulation layer. Since the materials of the insulation layer and conductor are fixed, both h and ρ are constants.

Based on the conductor structure shown in Figure 1, the following geometric relationships can be determined:

$$D_1 = D_2 + 2\Delta r, \tag{13}$$

$$D_2 = 2\sqrt{\frac{S_2}{\pi}}, \tag{14}$$

$$A_1 = 2L\sqrt{\pi}\sqrt{S_1 + S_2}. \tag{15}$$

Here, Δr is the thickness of the insulation layer.

Substituting Equations (13)–(15) into Equations (11) and (12) yields:

$$R_1 = \frac{k_{r1}}{\sqrt{S_1 + S_2}}, \tag{16}$$

$$R_2 = k_{r2} \cdot ln\left(1 + \frac{\Delta r}{\sqrt{\frac{S_2}{\pi}}}\right), \tag{17}$$

$$k_{r1} = \frac{1}{2hL\sqrt{\pi}}, \tag{18}$$

$$k_{r2} = \frac{\rho}{2\pi}. \tag{19}$$

Here, k_{r1} and k_{r2} are the constant coefficients for the calculation of the equivalent thermal resistances R_1 and R_2, respectively.

2.2.3. Calculation Equation for the Heat Transfer Coefficient of the Exposed Ends

Regarding the heat exchange coefficient k_1 for the exposed conductor sections at both ends, the exposed length is fixed at 150 mm. Consequently, k_1 is directly proportional to the surface area of the exposed conductor, which in turn is directly proportional to the diameter. Therefore, we have:

$$k_1 = k_k \cdot \sqrt{S_2}. \tag{20}$$

Here, k_k is the coefficient in the calculation formula for k_1. Since both the ambient environment and the conductor material are fixed, k_k is a constant.

2.2.4. Relationships Between Parameters and Conductor Specifications

The specification of the wire and cable conductor is defined as its cross-sectional area. Thus, we have:

$$S = S_2. \tag{21}$$

Here, S represents the specification of the wire and cable conductor.

According to the Chinese National Standard GB/T 5023.3-2008, there are eight types of conductors with specifications of 10 mm^2 or less. The correspondence between the conductor specification and the insulation layer thickness is presented in Table 1.

Table 1. Correspondence between insulation layer thickness and conductor specifications.

Conductor Specification (S/mm^2)	0.5	0.75	1.0	1.5	2.5	4.0	6.0	10.0
Insulation Layer Thickness (Δr/mm)	0.6	0.6	0.6	0.7	0.8	0.8	0.8	1.0

Data fitting techniques were employed to express the insulation layer thickness Δr as a mathematical function of the conductor specification S. By balancing fitting accuracy with computational efficiency, a hybrid model combining an exponential function and a quadratic polynomial was adopted. The fitting results are illustrated in Figure 3. The coefficient of determination (R^2) is 0.9738, and the resulting fitted equation is given by:

$$\Delta r(S) = 7389.0452 \times \left(1 - e^{-\frac{S}{78.513}}\right) + 0.5598 \cdot S^2 - 93.9113 \cdot S + 0.478. \tag{22}$$

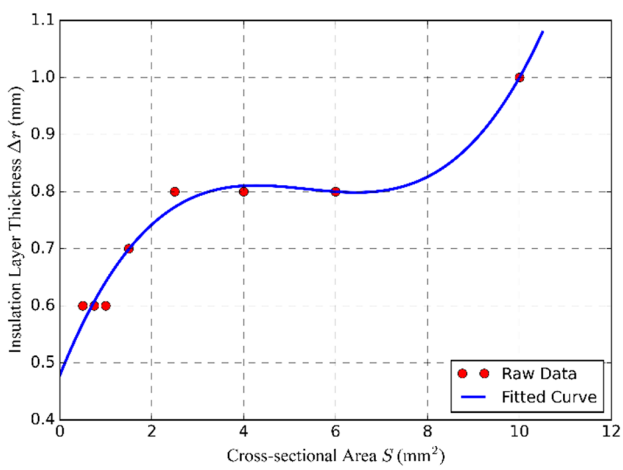


Figure 3. Fitting curve of insulation layer thickness versus conductor specification.

Based on the geometric structure of the wire and cable shown in Figure 3, the relationship between the cross-sectional area of the insulation layer S_1 and the conductor specification S is given by:

$$S_1 = \frac{\left(2\cdot\sqrt{\frac{S}{\pi}} + 2\Delta r(S)\right)^2}{4}\pi - S. \tag{23}$$

By substituting Equations (21) and (23) into Equations (7), (8), (16) and (17), the mathematical relationships between the heat transfer parameters (C_1, C_2, R_1, R_2, and k_1) and the conductor specification S are obtained as follows:

$$C_1(S) = k_{c1}\cdot\left(\frac{\left(2\cdot\sqrt{\frac{S}{\pi}} + 2\Delta r(S)\right)^2}{4}\pi - S\right), \tag{24}$$

$$C_2(S) = k_{c2}\cdot S, \tag{25}$$

$$R_1(S) = \frac{k_{r1}}{\sqrt{\frac{\left(2\cdot\sqrt{\frac{S}{\pi}}+2\Delta r(S)\right)^2}{4}\pi}}, \tag{26}$$

$$R_2(S) = k_{r2}\cdot ln\left(1 + \frac{\Delta r(S)}{\sqrt{\frac{S}{\pi}}}\right), \tag{27}$$

$$k_1(S) = k_k \cdot \sqrt{S}. \tag{28}$$

Equations (3), (4), (22) and (24)–(28) collectively constitute the universal analytical thermal circuit model. The coefficient vector and the heat transfer parameter vector of the model are defined as:

$$P = [k_{c1}, k_{r1}, k_{c2}, k_{r2}, k_k]^T, \tag{29}$$

$$\theta(S) = [R_1(S), C_1(S), R_2(S), C_2(S), k_1(S)]^T. \tag{30}$$

Here, P is the coefficient vector of the thermal circuit model, and $\theta(S)$ is the heat transfer parameter vector.

3. Solution of Thermal Circuit Model Coefficients

3.1. Ambient Temperature Rise and Fall Experiments

Experimental methods were employed to obtain the ambient temperature profiles during temperature rise and fall processes.

- Experimental Setup: The setup included a high-precision Pt100 platinum resistance temperature sensor (Huakong Electronic (Huizhou) Co., Ltd., Huizhou, China), a mercury thermometer, an isolated industrial-grade Pt100 temperature acquisition transmitter module, and a computer.
- Experimental Method:
  1. The wire and cable samples were positioned in accordance with the electrical performance test methods specified in the Chinese National Standard GB/T 3048.4-2007.
  2. The high-precision Pt100 sensor was placed in the vicinity of the samples.
  3. To ensure measurement accuracy, the readings of the Pt100 sensor were first calibrated against a mercury thermometer to verify its precision. Subsequently,

the Pt100 sensor was positioned as shown in Figure 4 to measure the ambient temperature in real time.

4. To simulate the cable's response under different thermal dynamic processes, an air conditioning system was utilized to dynamically regulate the ambient temperature. The specific regulation protocols were as follows: For the temperature rise process, the air conditioner was first activated to stabilize the ambient temperature at 16 °C, and then switched off to allow the temperature to rise naturally. For the temperature fall process, the air conditioner was first set to stabilize the temperature at 26 °C, and then switched to cooling mode set at 20 °C to lower the ambient temperature.
5. Data recording was performed via the computer at an interval of 1 s.

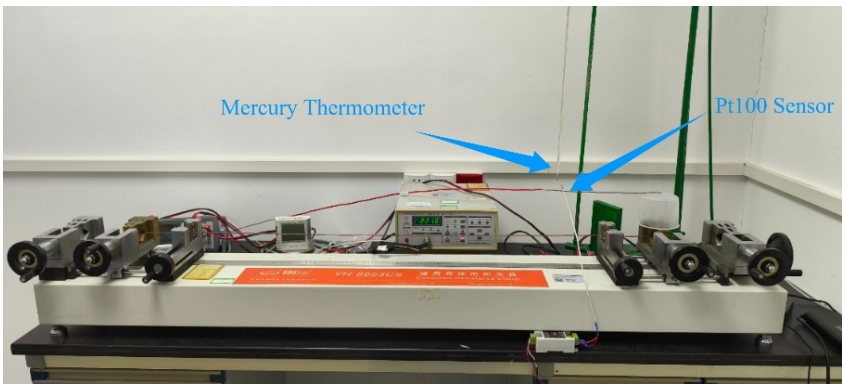
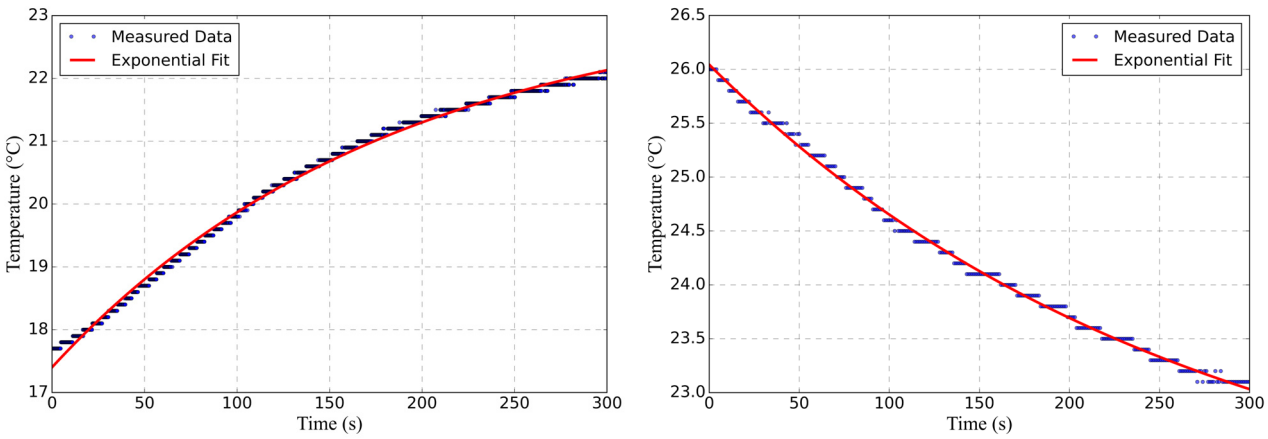


Figure 4. Placement of the temperature sensors.

The experimental results are presented in Figure 5. It can be observed that the curves for both the ambient temperature rise and fall closely resemble exponential waveforms. Therefore, the nonlinear least squares method was employed to fit the experimental data using an exponential function. The fitted equations for the temperature rise and fall are given by Equations (31) and (32), respectively. The coefficients of determination (R^2) for the fits are 0.9969 and 0.9973, respectively.

$$T(t) = 23.28 - 5.88e^{-t/183.90}, \tag{31}$$

$$T(t) = 21.57 + 4.47e^{-t/268.44}. \tag{32}$$



(**a**) Temperature rise (**b**) Temperature fall

Figure 5. Measurement results and fitted curves of the ambient temperature.

3.2. Finite Element Simulation

Based on the ambient temperature profiles obtained in Section 3.1, finite element software was employed to simulate the corresponding conductor temperature variations. Simulations were conducted for all eight specifications of wire and cable conductors listed in Table 1.

3.2.1. Geometric Modeling

Based on the wire and cable structure shown in Figure 1 and the dimensions listed in Table 1, a geometric simulation model of the wire and cable was established. In this model, the length of the insulation layer was set to 1m, and the length of the exposed conductor sections at each end was set to 150 mm. The constructed geometric model of the cable is illustrated in Figure 6.

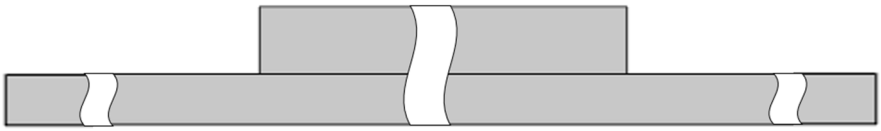


Figure 6. Geometric model of the electric wire and cable.

3.2.2. Definition of Physical Domains

The material for the conductor was set to copper, and the material for the insulation layer was set to polyvinyl chloride (PVC). The properties of these two materials are listed in Table 2.

Table 2. Material properties.

Material	Thermal Conductivity k (W/(m·K))	Density ρ (kg/m^3)	Specific Heat Capacity C_p (J/(kg K))	Temperature Coefficient of Resistance α (1/K)
Copper	400	8960	385	0.00393
Polyvinyl Chloride	0.16	1380	900	—

The external air environment surrounding the conductor was defined as a fluid domain. To simulate the thermal influence of the external environment on the conductor, a heat flux was applied to all outer geometric boundaries of the conductor, with the combined heat transfer coefficient h set to 10 W/(m^2·K).

3.2.3. Mesh Generation

Given that the wire and cable conductor possesses a simple structure characterized by a regular coaxial cylindrical geometry, the mesh was generated using the software's preset "Extra fine" element size. Taking the 2.5 mm^2 specification as an example, the mesh configuration is illustrated in Figure 7, which comprises 4081 domain elements and 3039 boundary elements.

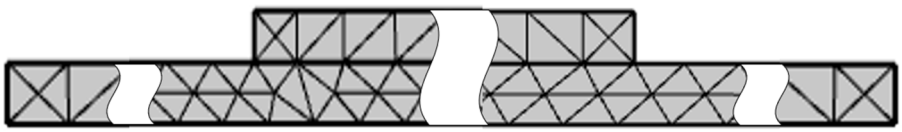


Figure 7. Finite element mesh of the electric wire and cable.

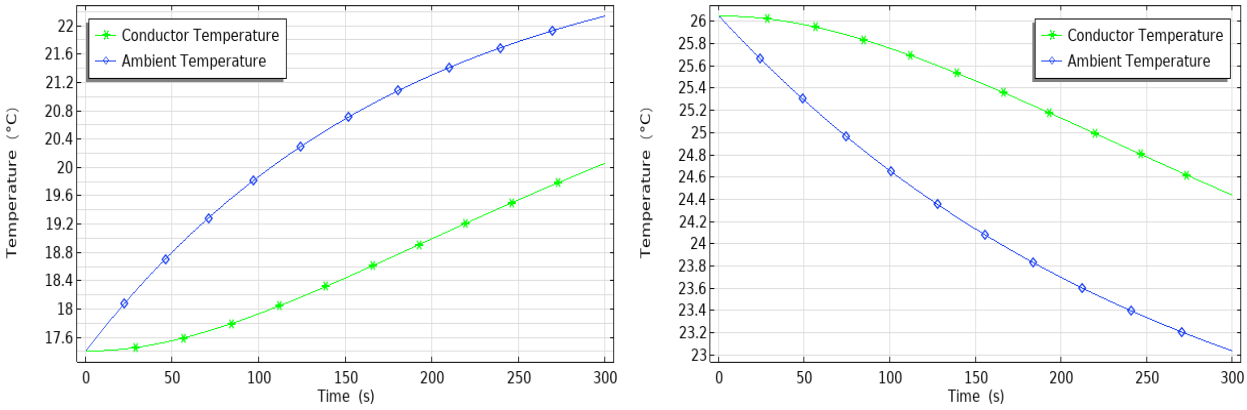
3.2.4. Simulation Parameters and Solver Settings

The initial temperatures of the conductor and the insulation layer were set to match the initial ambient temperature. A transient solver was employed for the calculation. The external ambient temperature variations were configured according to Equations (31) and (32),

with a simulation duration of 300 s for both the temperature rise and temperature fall processes.

3.2.5. Simulation Results

Simulation outputs yielded conductor temperature curves for all eight solid core cable specifications detailed in Table 2. As a representative example, the simulation results for the 2.5 mm^2 conductor are depicted in Figure 8.



(**a**) Temperature rise (**b**) Temperature fall

Figure 8. Simulation results for the 2.5 mm^2 conductor.

3.3. Coefficient Solution Algorithm

3.3.1. Determination of Constant Coefficients

Since the thermal capacitance depends on the mass and specific heat capacity of the heat-conducting material, and considering that the mass remains constant during the cable temperature rise process while the variation in specific heat capacity is negligible within a certain temperature range, k_{c1} and k_{c2} can be treated as constant coefficients. By substituting the material parameters from Table 2 into Equations (9) and (10), the calculated values for k_{c1} and k_{c2} are obtained and listed in Table 3.

Table 3. Values of the constant coefficients k_{c1} and k_{c2}.

k_{c1}	k_{c2}
1.2420	3.4496

3.3.2. Determination of Dynamic Coefficients via Differential Evolution

The calibrated values of the combined heat transfer coefficient h and the thermal resistance coefficient ρ in the thermal resistance equations are related to operating conditions [16]. Research by Hu et al. [19] indicates that ρ changes dynamically with temperature, and significant differences exist when temperature gaps are large. Therefore, h and ρ are likely dependent on specific application conditions. References [20,21] utilized finite element simulation results to correct the parameters of thermal circuit models. Following these methods, this study utilizes the ambient and conductor temperature profiles obtained in Sections 3.1 and 3.2 and employs the Differential Evolution (DE) algorithm to solve for k_{r1}, k_{r2}, and k_k. The DE algorithm was selected for parameter identification due to its superior performance in continuous parameter optimization spaces. Compared to other common evolutionary optimizers such as Genetic Algorithms or Particle Swarm Optimization, DE demonstrates faster convergence, fewer hyper-parameters, and better robustness to noisy fitness landscapes [22,23]. These characteristics make it highly suitable for the continuous

heat-transfer coefficient inversion and the non-convex objective function derived from the FEM-based temperature curves in this study. The algorithm workflow is as follows:

1. Initialization: The search boundaries were set as $k_{r1} \in [10, 60]$, $k_{r2} \in [0.5, 1.5]$, and $k_k \in [0, 5]$. An initial population was randomly generated within these ranges, where each individual in the population is a three-dimensional vector $[k_{r1}, k_{r2}, k_k]^T$
2. Iteration: Through mutation, crossover, and selection operations, the population was iterated to identify a set of optimal coefficients that minimize the Root Mean Square Error (RMSE) between the calculated values of the thermal circuit model and the finite element simulation values.
3. Validation: To evaluate the generalization ability of the model, Leave-One-Out Cross-Validation (LOOCV) was employed. Specifically, 8 rounds of validation were conducted. In each round, the conductor temperature data of the i-th specification ($i = 1$–8) served as the validation set, while data from the other 7 specifications served as the training set.
4. Finalization: After confirming the validity of the modeling method through cross-validation, the data from all 8 specifications were aggregated into a complete training set. The resulting values for k_{r1}, k_{r2}, and k_k are listed Table 4, and the LOOCV results are presented in Table 5.

Table 4. Values of the optimized coefficients k_{r1}, k_{r2}, and k_k.

k_{r1}	k_{r2}	k_k
53.4903	1.5	0.0138

Table 5. Error analysis metrics from the cross-validation.

Average Generalization RMSE (°C)	Standard Deviation (°C)
0.0224	0.0184

The identified coefficients possess specific physical interpretations derived from the governing equations. Specifically, k_{c1} and k_{c2} encapsulate the volumetric heat capacities of the insulation and conductor, determined by their respective densities (ρ) and specific heat capacities (c_p). k_{r2} reflects the thermal resistivity (ρ) of the insulation material, while k_{r1} and k_k characterize the convective heat transfer capabilities at the cable surface and exposed ends, respectively. Substituting the determined coefficient values into Equations (3), (4) and (24)–(28) yields the derived universal analytical thermal circuit model for civil cables:

$$\frac{dT_{ins}}{dt} = \frac{T_{env} - T_{ins}}{R_1(S)C_1(S)} - \frac{T_{ins} - T_{core}}{R_2(S)C_1(S)}, \tag{33}$$

$$\frac{dT_{core}}{dt} = \frac{T_{ins} - T_{core}}{R_2(S)C_2(S)} + \frac{k_1(S)(T_{env} - T_{core})}{C_2(S)}, \tag{34}$$

$$C_1(S) = 1.242\left(\frac{\left(2\cdot\sqrt{\frac{S}{\pi}} + 2\Delta r(S)\right)^2}{4}\pi - S\right), \tag{35}$$

$$C_2(S) = 3.4496\cdot S, \tag{36}$$

$$R_1(S) = \frac{53.4903}{\sqrt{\frac{\left(2\cdot\sqrt{\frac{S}{\pi}}+2\Delta r(S)\right)^2}{4}\pi}}, \tag{37}$$

$$R_2(S) = 1.5 \cdot ln\left(1 + \frac{\Delta r(S)}{\sqrt{\frac{S}{\pi}}}\right), \tag{38}$$

$$k_1(S) = 0.0138 \cdot \sqrt{S}. \tag{39}$$

Here, $\Delta r(S)$ is calculated according to Equation (22).

4. Test of Model Estimation Accuracy

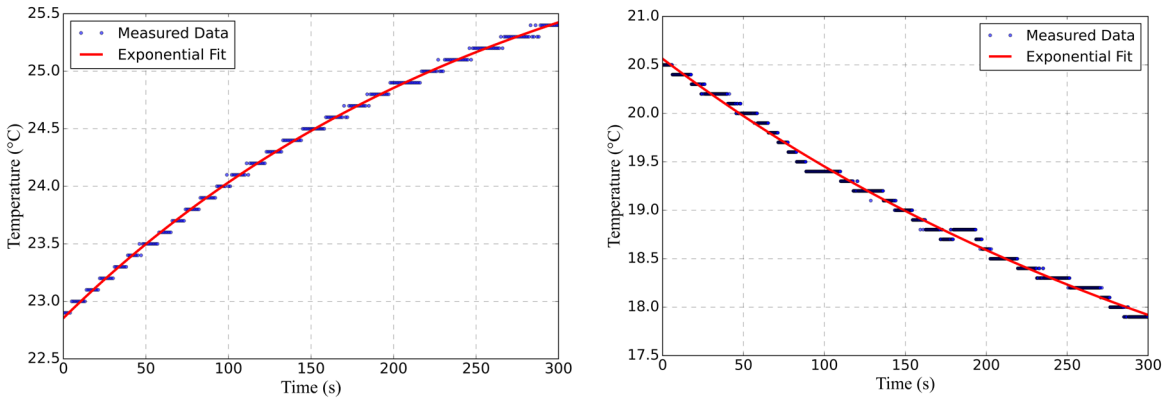
The temperature profile data shown in Figure 8 served as the training set for solving the model coefficients. To validate the generalization ability and computational accuracy of the model, temperature profile data under different operating conditions were employed to test the model's estimation accuracy.

Differences between Testing and Training Conditions:

1. Different Ambient Temperature Variations. During testing, the ambient temperature rise and fall were regulated as follows: For the temperature rise process, the air conditioner was first activated to stabilize the ambient temperature at approximately 22 °C, and then the setpoint was adjusted to 28 °C to induce a temperature rise. For the temperature fall process, the air conditioner was first activated to stabilize the ambient temperature at approximately 22 °C, and then the setpoint was adjusted to 16 °C to induce a temperature drop. The ambient temperature was measured using the same experimental method described in Section 3.1. The measured ambient temperature variation data and their corresponding fitted curves are illustrated in Figure 9. Equations (40) and (41) represent the fitted equations for the temperature rise and temperature fall curves, respectively.

$$T_{env}(t) = 26.73 - 3.88e^{-t/275.64}, \tag{40}$$

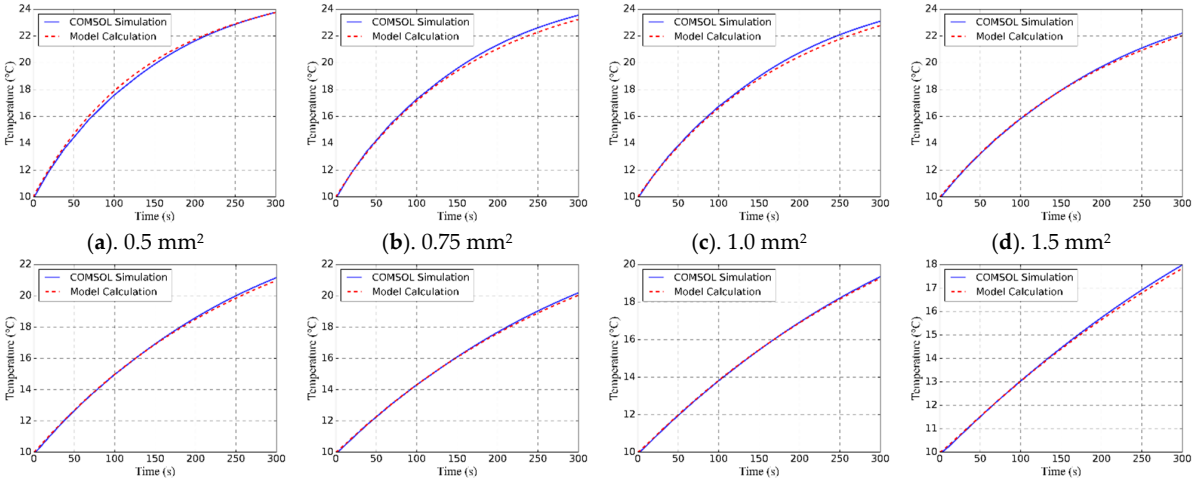
$$T_{env}(t) = 15.60 + 4.96e^{-t/394.04}. \tag{41}$$



(**a**) Temperature rise (**b**) Temperature fall

Figure 9. Ambient temperature variations under testing conditions.

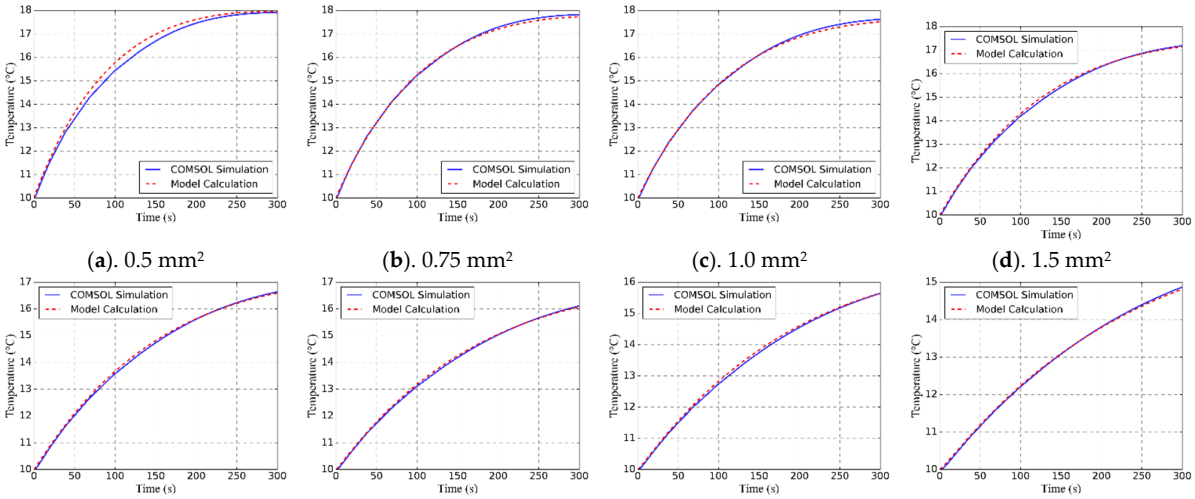
2. Different Initial Conductor Temperatures. To simulate an autumn or winter scenario, the outdoor temperature was assumed to be 10 °C, and the initial temperature of the conductor was set to match this outdoor temperature. Using the same finite element simulation method described in Section 3.2 and based on the ambient temperature profiles under testing conditions shown in Figure 9, the corresponding conductor temperature profiles for various specifications were obtained. These are illustrated in Figures 10 and 11, with the corresponding errors shown in Figures 12 and 13.



(**a**). 0.5 mm² (**b**). 0.75 mm² (**c**). 1.0 mm² (**d**). 1.5 mm²

(**e**). 2.5 mm² (**f**). 4.0 mm² (**g**). 6.0 mm² (**h**). 10.0 mm²

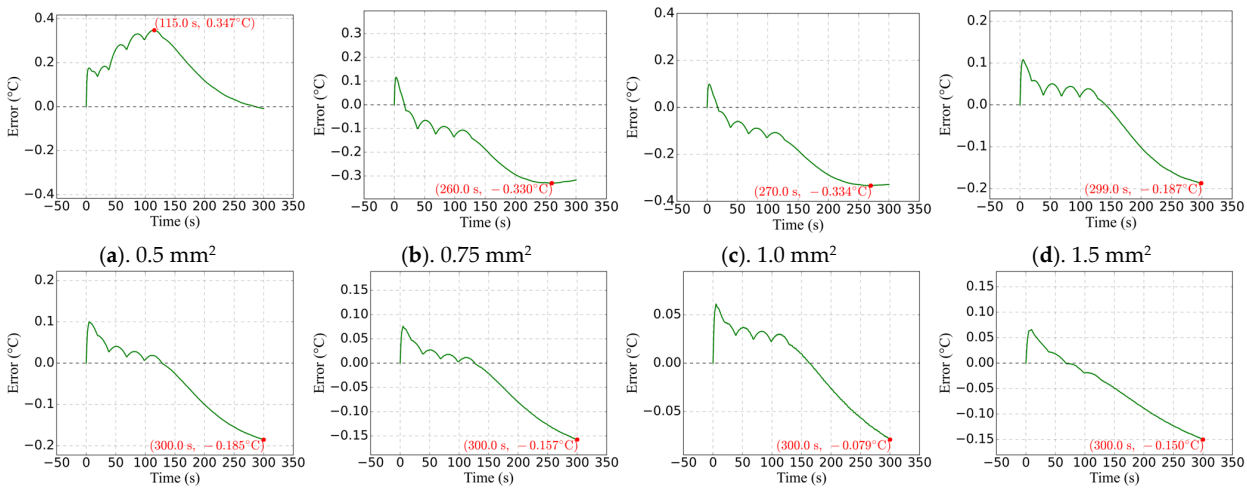
Figure 10. Temperature errors between the model calculation and simulation results under ambient temperature rise conditions.



(**a**). 0.5 mm² (**b**). 0.75 mm² (**c**). 1.0 mm² (**d**). 1.5 mm²

(**e**). 2.5 mm² (**f**). 4.0 mm² (**g**). 6.0 mm² (**h**). 10.0 mm²

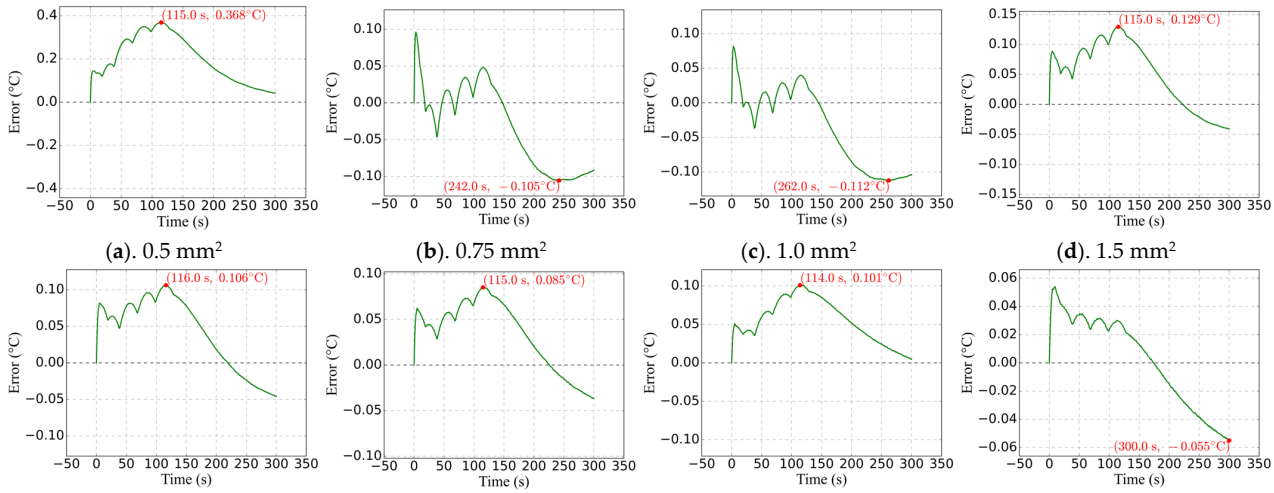
Figure 11. Comparison between calculated and simulated results of conductor temperature with ambient temperature falling.



(**a**). 0.5 mm² (**b**). 0.75 mm² (**c**). 1.0 mm² (**d**). 1.5 mm²

(**e**). 2.5 mm² (**f**). 4.0 mm² (**g**). 6.0 mm² (**h**). 10.0 mm²

Figure 12. Temperature errors between calculated and simulated results with ambient temperature rising.



(**a**). 0.5 mm^2 (**b**). 0.75 mm^2 (**c**). 1.0 mm^2 (**d**). 1.5 mm^2

(**e**). 2.5 mm^2 (**f**). 4.0 mm^2 (**g**). 6.0 mm^2 (**h**). 10.0 mm^2

Figure 13. Temperature errors between the model calculation and simulation results under ambient temperature fall conditions.

The ambient temperature data under testing conditions and the initial conductor temperature were input into the universal analytical thermal circuit model to calculate the conductor temperature profiles for the various specifications, as plotted in Figures 10 and 12.

A comparison between the conductor temperatures calculated by the thermal circuit model and those obtained from finite element simulation reveals that the two are in close agreement. The maximum deviation occurred in the 0.5 mm^2 specification. Specifically, during the temperature rise phase, at an ambient temperature of 24.174 °C, the simulated conductor temperature was 18.311 °C, resulting in a model calculation deviation of 0.347 °C. During the temperature fall phase, at an ambient temperature of 19.305 °C, the simulated conductor temperature was 15.843 °C, resulting in a calculation deviation of 0.368 °C.

$$R_{20} = R_t \frac{1}{1 + 0.00393(t - 20)}. \tag{42}$$

Here, R_t is the resistance value of the conductor at temperature t, and t is the conductor temperature.

Equation (42) represents the formula for converting the real-time resistance R_t at temperature t to R_{20}. Based on this equation, the relative errors of the R_{20} value under the aforementioned maximum temperature deviation conditions are calculated to be 0.147% and 0.148%, respectively. These values are far below the 0.5% accuracy threshold required by the standard for conductor resistance testing. Furthermore, by applying the thermal circuit model, the conductor temperature can be calculated within a very short duration after the initial moment. As indicated in Figures 11 and 13, the conductor temperature deviation at this early stage is significantly smaller than the maximum deviation, implying that the relative error of the R_{20} value in actual measurement scenarios will be significantly less than 0.147% and 0.148%.

5. Discussion and Conclusions

5.1. Discussion

In the model training process described in Section 3, the specific operating condition employed involved an ambient temperature rise from 16 °C to 20 °C and a drop from 26 °C to 20 °C, with the initial conductor temperature set equal to the initial ambient temperature. Although this represents a specific case among the myriad possibilities encountered in practice, training the model using this specific case does not compromise the universality of

the results. The rationale is that the model coefficients characterize the inherent heat transfer properties of the wire and cable, which are independent of temperature variations. While different operating conditions yield different simulated conductor temperature profiles, the underlying heat transfer characteristics derived from any combination of operating conditions and their corresponding thermal responses remain consistent. This consistency is further corroborated by the testing results under different conditions presented in Section 4.

In the standard measurement of the R_{20} resistance for civil wire and cable conductors, the sample is typically conditioned in a 20 °C environment for a sufficient period to ensure the conductor temperature stabilizes at 20 °C before direct resistance measurement. Based on the conductor thermal circuit model established in this study, numerical calculation methods can be further developed to compute the real-time conductor temperature. This approach effectively addresses the challenge of direct conductor temperature measurement, thereby eliminating the need for this prolonged waiting period. Consequently, the real-time resistance of the conductor can be measured immediately and converted to the standard resistance.

5.2. Conclusions

1. During the solution process for the thermal circuit model coefficients, the average generalization Root Mean Square Error (RMSE) and standard deviation for wire and cable conductors of various specifications were found to be less than 0.0224 °C and 0.0184 °C, respectively. This indicates that the solutions for the model coefficients exhibit excellent stability and consistency under different training samples.
2. Test results demonstrate that the maximum temperature calculation deviation of the thermal circuit model is consistently less than 0.368 °C, and the relative measurement error for the R_{20} value is less than 0.148%. This signifies that the established thermal circuit model possesses high calculation accuracy, enabling the acquisition of precise values for conductor resistance.
3. Satisfactory test results were achieved even when there was a significant disparity between the initial conductor temperature and the initial ambient temperature in the test scenarios. This suggests that the established thermal circuit model demonstrates strong robustness.
4. The test results show that the thermal circuit model maintains high calculation accuracy under various specifications of cables. This confirms that the established model has the universality fitting for application to various specifications of civil electric cables.

Author Contributions: Conceptualization, C.L. and Z.Z.; methodology, C.L., K.M. and H.L.; software, K.M.; validation, K.M., N.Y. and H.L.; formal analysis, C.L. and K.M.; investigation, K.M.; resources, Z.Z.; data curation, K.M.; writing—original draft preparation, K.M. and N.Y.; writing—review and editing, C.L., H.L. and Z.Z.; visualization, K.M.; supervision, H.L. and Z.Z.; project administration, C.L.; funding acquisition, C.L. All authors have read and agreed to the published version of the manuscript.

Funding: This research was supported by "The Guangdong Innovation Design and Manufacturing Engineering Technology Research Center for Small Household Appliances".

Data Availability Statement: The data that support the findings of this study are available on request from the corresponding author.

Acknowledgments: The authors acknowledge "Guangdong Engineering Technology Research Center for the Innovative Design and Manufacturing of Small Household Electrical Appliances" for supports.

Conflicts of Interest: The authors declare no conflict of interest.

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